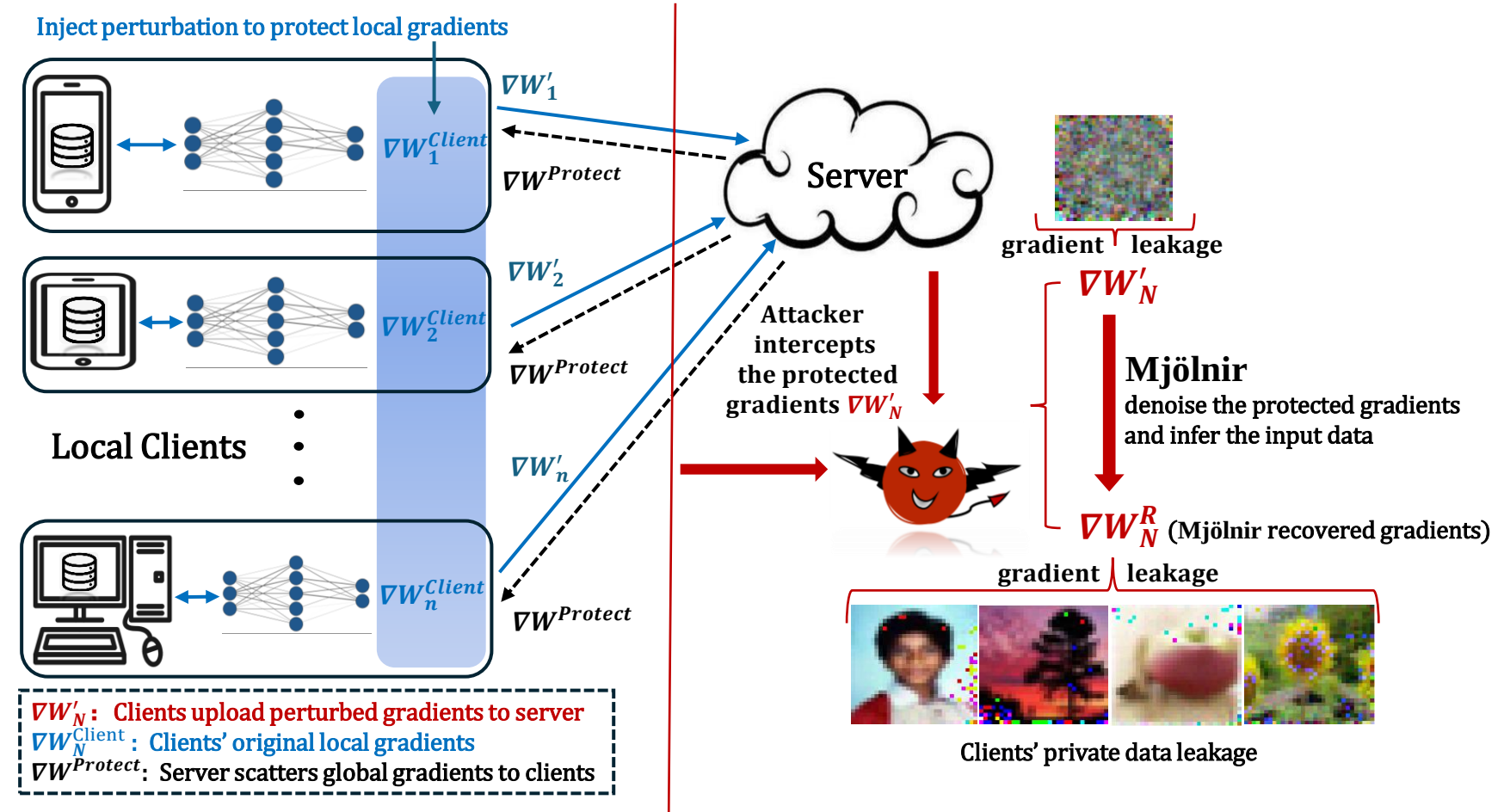

Inject perturbation to protect local gradients
∇W_1^{Client}
∇W_2^{Client}
∇W_n^{Client}
Local Clients
$\nabla W'_1$
$\nabla W^{Protect}$
$\nabla W'_2$
$\nabla W^{Protect}$
$\nabla W'_n$
$\nabla W^{Protect}$
Server
gradient leakage
$\nabla W'_N$
Attacker intercepts the protected gradients $\nabla W'_N$
Mjölnir
denoise the protected gradients and infer the input data
∇W_N^R (Mjölnir recovered gradients)
gradient leakage
Clients' private data leakage
$\nabla W'_N$: Clients upload perturbed gradients to server
∇W_N^{Client}: Clients' original local gradients
$\nabla W^{Protect}$: Server scatters global gradients to clients